Presented by
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SOFTWARE ENGINEERING

M01 | S02: Software Life Cycles, Feasibility & Risk

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2 What Is a Software Process?

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MOODLE CHECKPOINTS**

Submission	Required coverage	Points
Checkpoint 1	M01–M02	20
Checkpoint 2	M01–M04	20
Checkpoint 3	M01–M06	20
Checkpoint 4	M01–M08	20
Checkpoint 5 — final project	M01–M10	20
Total	**10 modules**	**100**

Checkpoint 1
M01–M02

Checkpoint 2
M01–M04

Checkpoint 3
M01–M06

Checkpoint 4
M01–M08

Checkpoint 5
final project*

*final project submission - 1 week after our last class

*Always export your project before leaving the lab. A Moodle checkpoint is not the same thing as a progress backup.

Syllabus topic	Coverage
Project vision, lifecycle and feasibility	M01
Business, functional and non-functional requirements	M02
System functions and information flows	M03–M04
Reliability, availability, security and fault tolerance	M05
Physical infrastructure and deployment	M06
UML diagrams and design patterns	M03, M07–M08
CASE tools	Website modelling tools used in required diagram tasks
Project management and professional practice	M09
Testing methods and practical exercises	M10: test plan, six test cases and two manual walkthroughs

QuickBite project - criteria / requirements

1. **Work in teams of 3**

Submit one agreed project ZIP per team. Everyone should contribute and be able to explain the team's decisions.

2. **Each module is worth 10 points**

Each module has 5 criteria worth 0–2 points each, published with its instructions:

- 2 points: complete, clear and justified.
- 1 point: partly correct, but incomplete or unclear.
- 0 points: missing or fundamentally incorrect.

3. **Different answers can earn full marks**

We assess your reasoning, use of the project brief and consistency between your answers. There is no single ideal solution.

4. **Focus on required tasks**

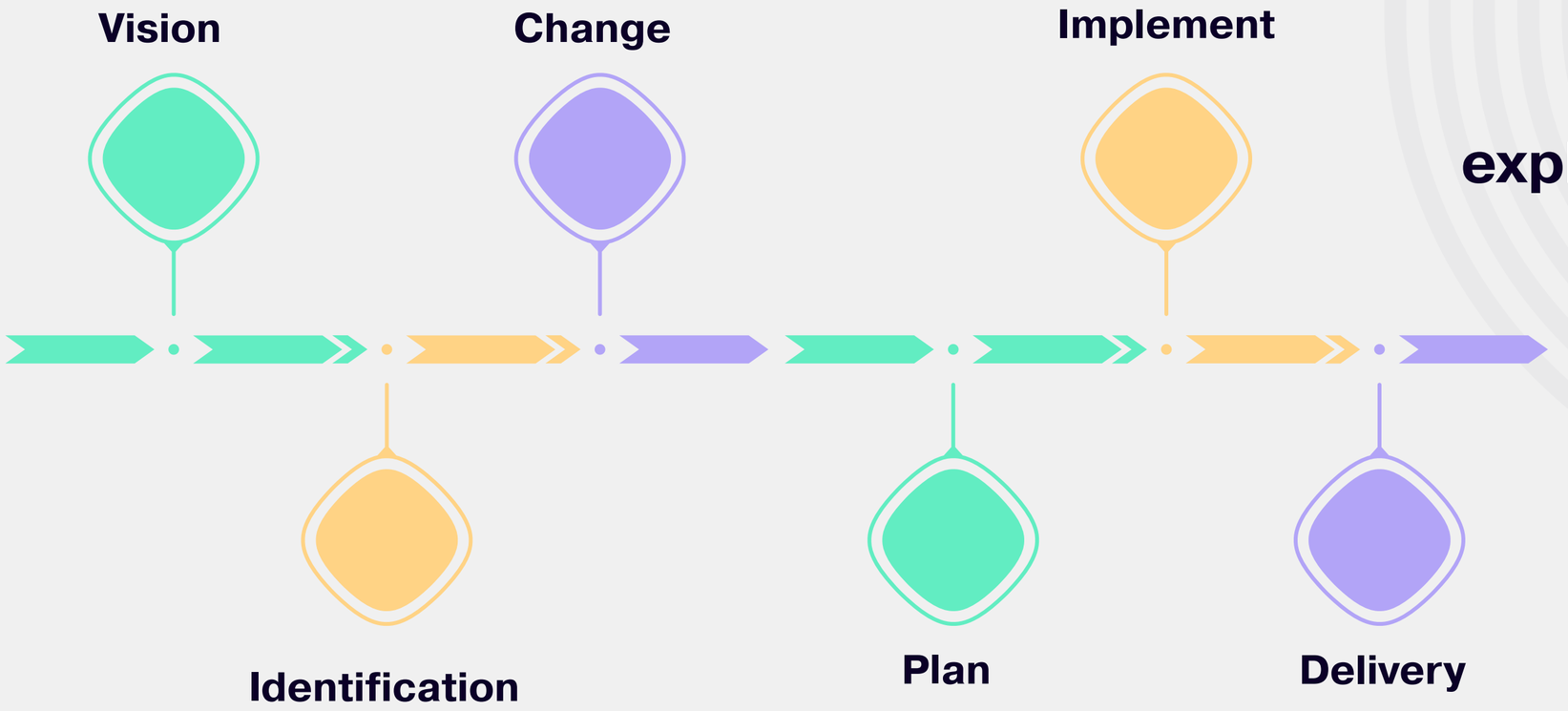
Optional tasks are extra practice. Skipping them causes no penalty; completing them adds no points.

5. **Submit your work through Moodle**

Save your answers on the website, export the latest project ZIP and upload it by the assignment deadline. "Saved" on the website confirms storage - it is not a grade.

6. **Project total: 100 points**

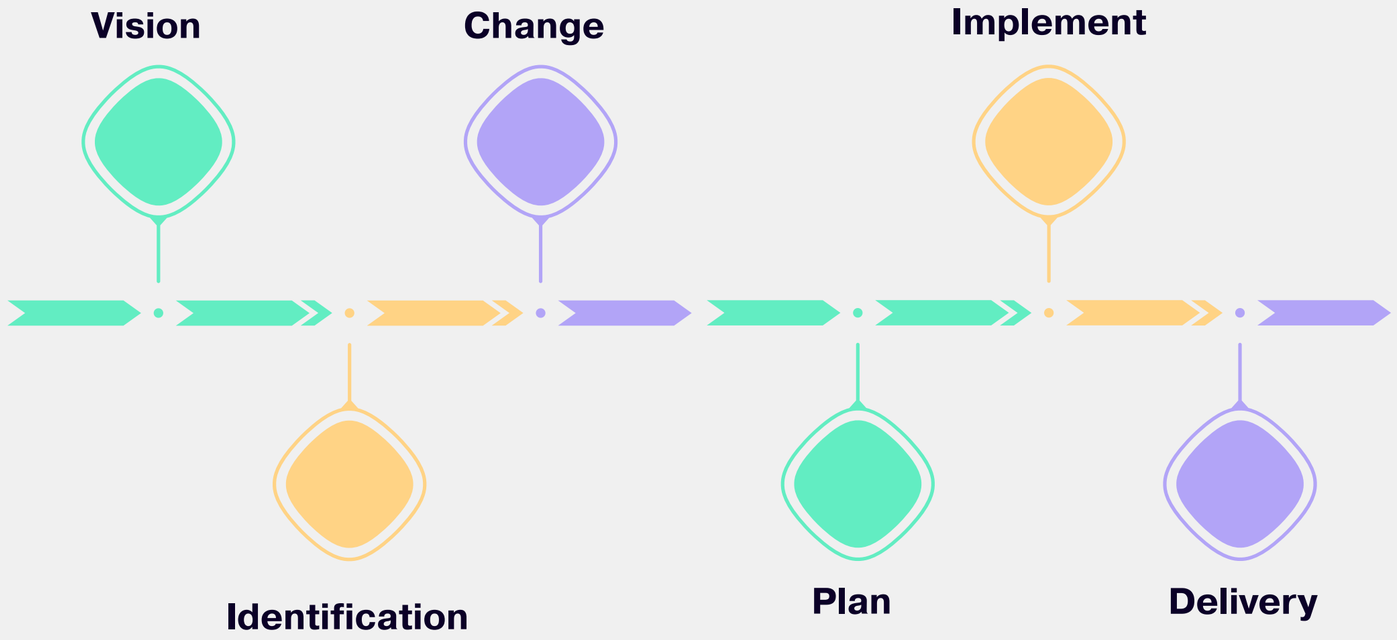
M01 is worth 10 points across Sessions 1 and 2 together. The project contributes to the laboratory component;



A project vision explains why change is needed.

A software process explains how we organize the work needed to deliver that change.

FROM VISION TO DELIVERY



A project vision explains why change is needed.

A software process explains how we organize the work needed to deliver that change.

FROM VISION TO DELIVERY

Car Rental example

Vision → reduce booking conflicts and manual work.

Process → decide how requirements, design, testing and delivery will be organized.

WHAT IS A SOFTWARE PROCESS?

A software process is a structured set of activities used to develop and evolve software.



Typical activities:

- specification,
- design & implementation,
- validation,
- evolution.

WHAT IS A SOFTWARE PROCESS? - CAR RENTAL EXAMPLE

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Car Rental example:

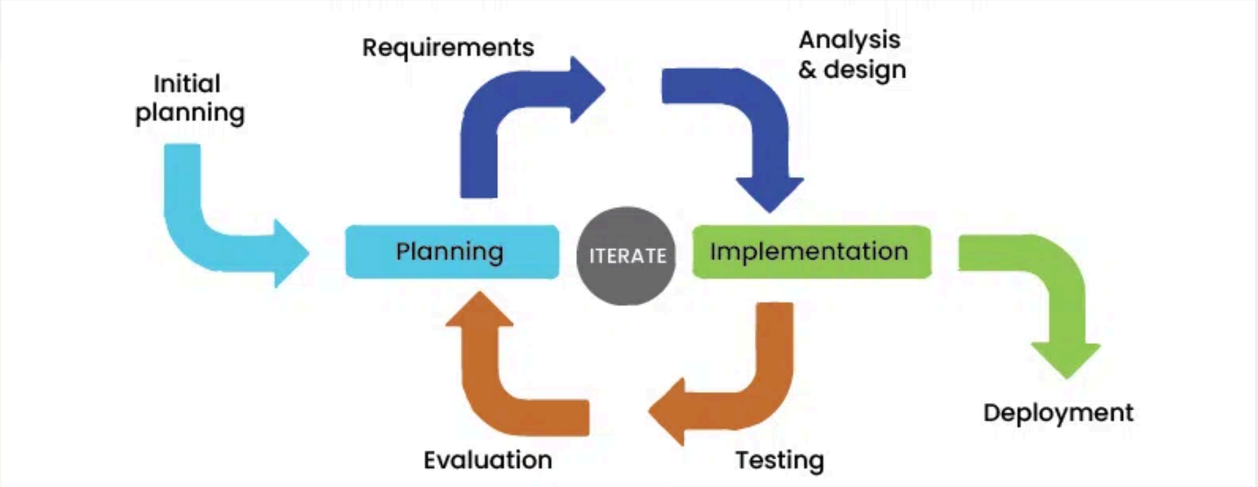
define booking rules → design reservation flow → build → test → improve

WHAT IS A PROCESS MODEL?

A process model is a simplified representation of how development is organized.

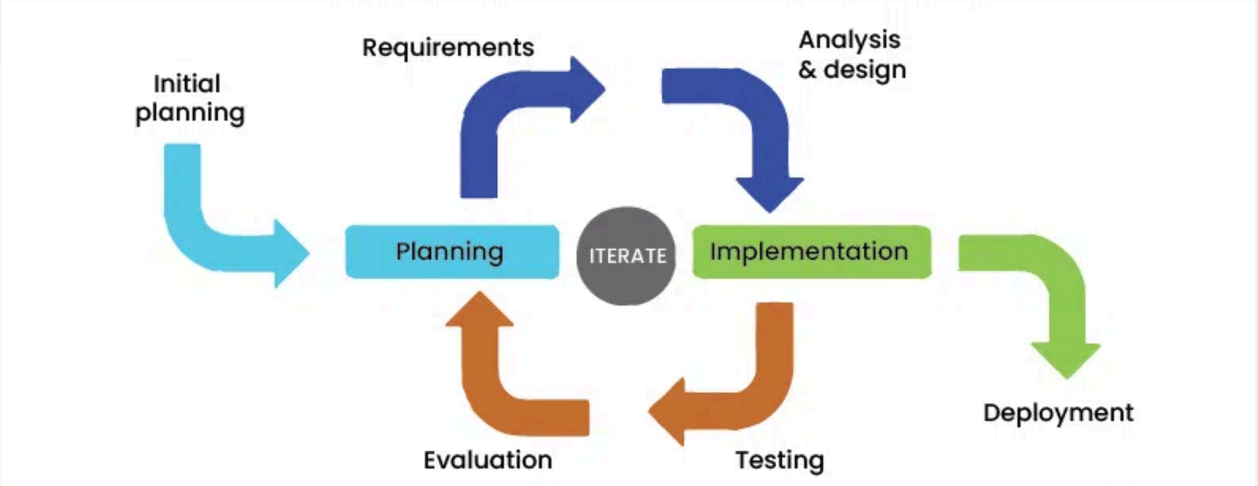
It helps answer:

- What happens first?
- When do we get feedback?
- What can change?
- When do we deliver?



WHAT IS A PROCESS MODEL? - CAR RENTAL EXAMPLE

A process model is a simplified representation of how development is organized.



It helps answer:

- What happens first?
- When do we get feedback?
- What can change?
- When do we deliver?

Car Rental example:

Do we define everything first, or test booking ideas with branch staff early?

Sequential / Waterfall Model - Car Rental example

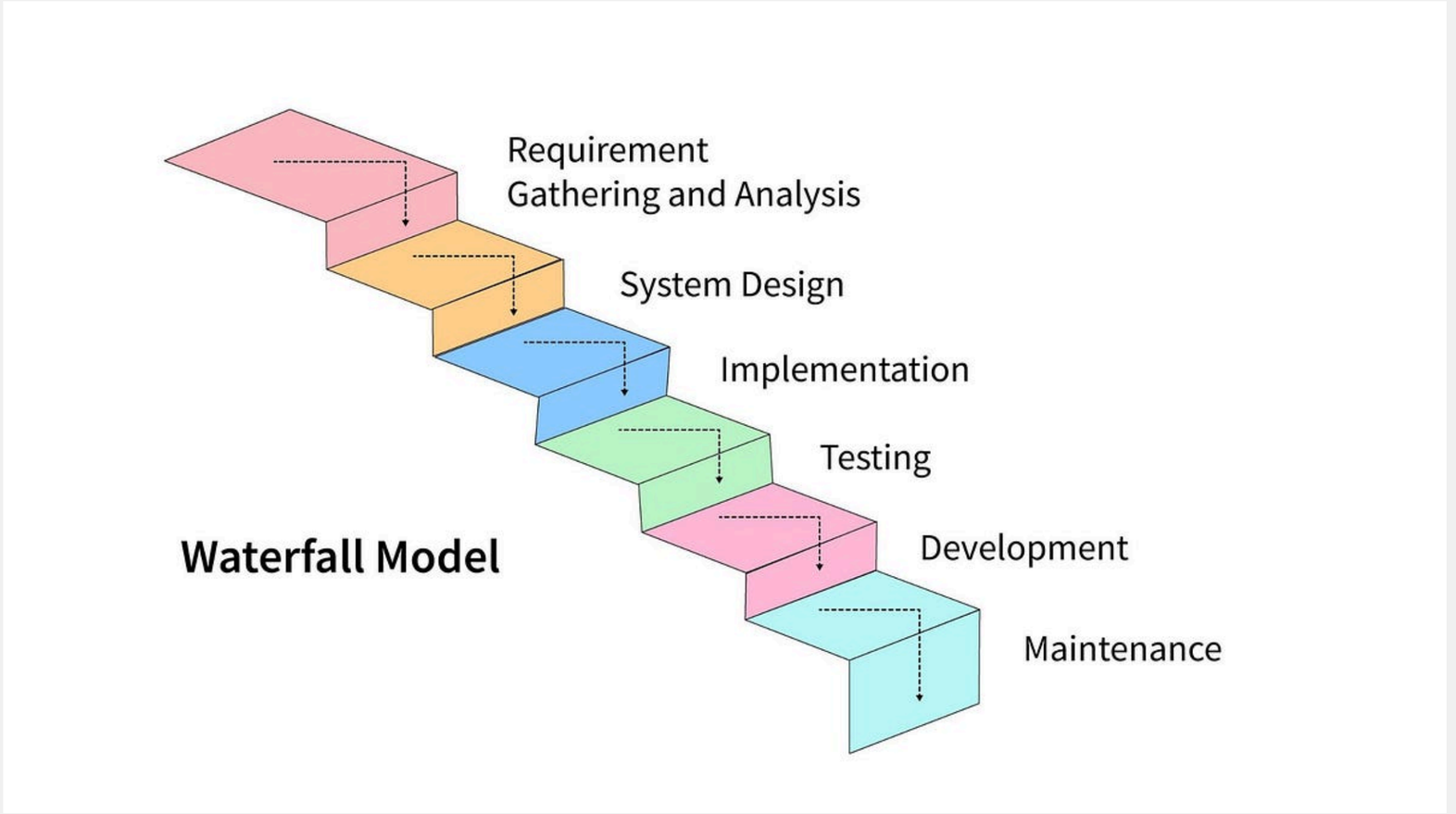
Car Rental example:
All reservation rules are agreed before implementation begins.

Useful when:

- requirements are **stable**,
- change is **limited**,
- **documentation** is important.

Waterfall Model is a traditional software development approach that follows a **linear, step-by-step process** where **each phase is completed before moving to the next**.

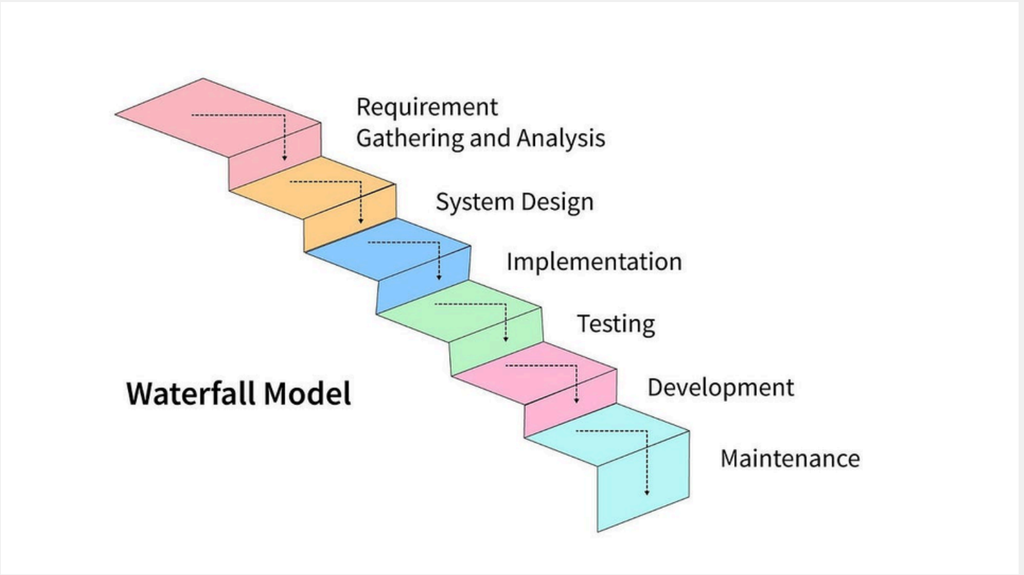
Waterfall **locks** specifications down via rigid phase gates with **little room for mid-project adjustments.**



Sequential / Waterfall Model - Car Rental example

Useful when:

- requirements are **stable**,
- change is **limited**,
- **documentation** is important.



Car Rental example:

1. Define all booking requirements.
2. Design reservation and fleet logic.
3. Implement.
4. Test.
5. Deploy.

Risk:

A major missing requirement may be discovered very late.

Agile vs. Plan-driven

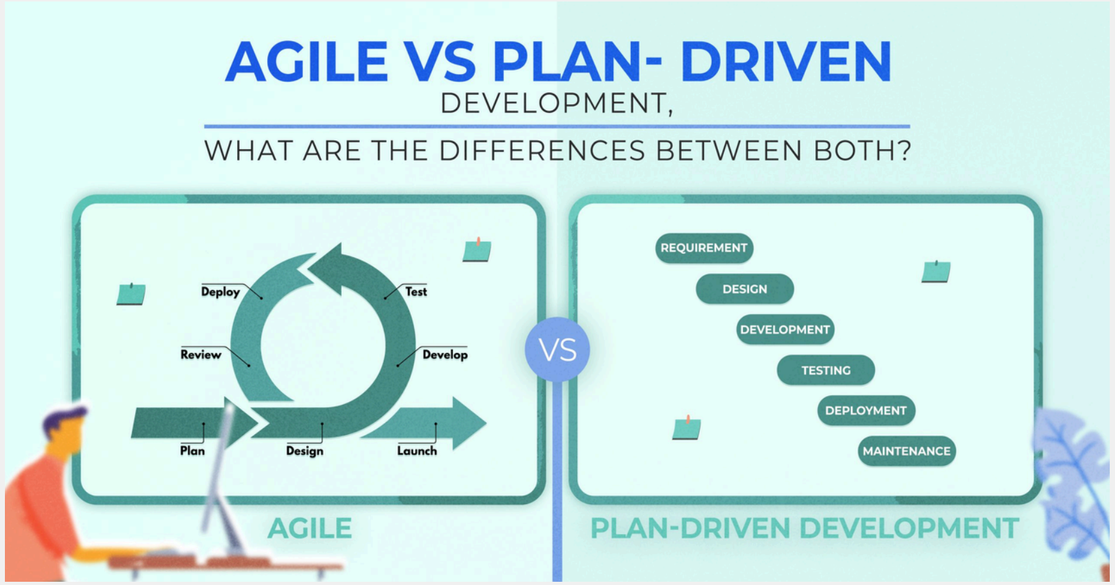
Agile

- shorter cycles,
- frequent feedback,
- easier response to change.

Plan-driven

- more upfront planning,
- stronger documentation,
- predictable phases.

The Agile Model is an **iterative and incremental** software development approach designed to deliver software **quickly** while **adapting** to **changing customer requirements** through continuous collaboration and feedback



Plan-driven development is a broad software engineering approach where all activities - such as requirements, design, scheduling, and verification - are **thoroughly planned and documented before coding begins, <u>allowing for mid-project adjustmens</u>.**

Can the approaches be mixed?

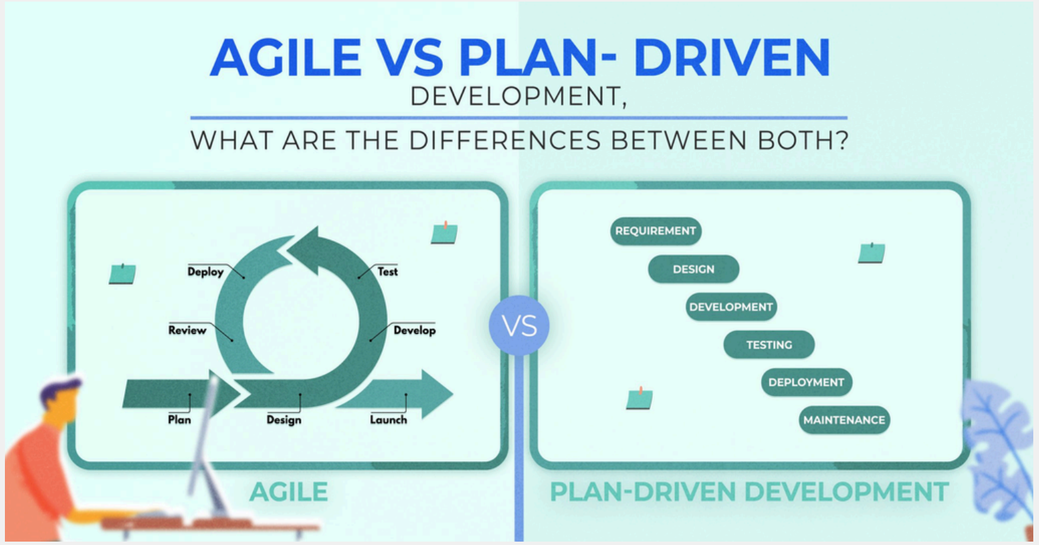
Agile vs. Plan-driven - Car Rental example

Agile

- shorter cycles,
- frequent **feedback**,
- easier response to change.

Plan-driven

- more **upfront** planning,
- stronger documentation,
- predictable phases.



Can the approaches be mixed? Sure!

Car Rental example

Stable legal rules may be planned **upfront**, while booking UX can evolve through **feedback**.

Agile vs. Waterfall

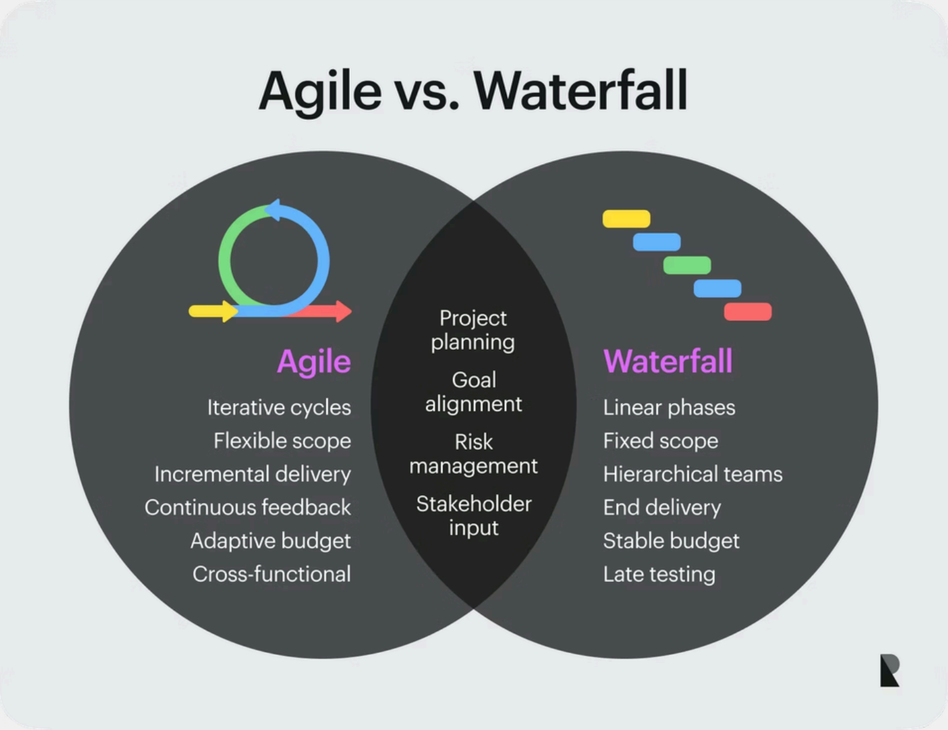
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Waterfall Model is a **traditional** software development approach that follows a l**inear, step-by-step** process where each phase is completed before moving to the next. Less room for mid-project adjustments

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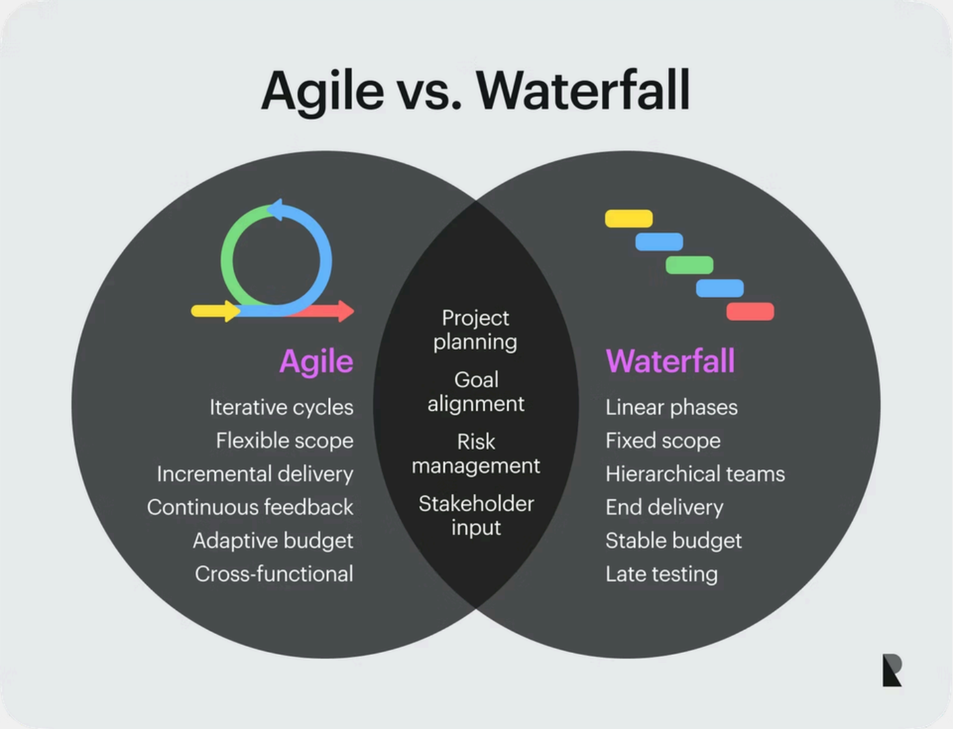
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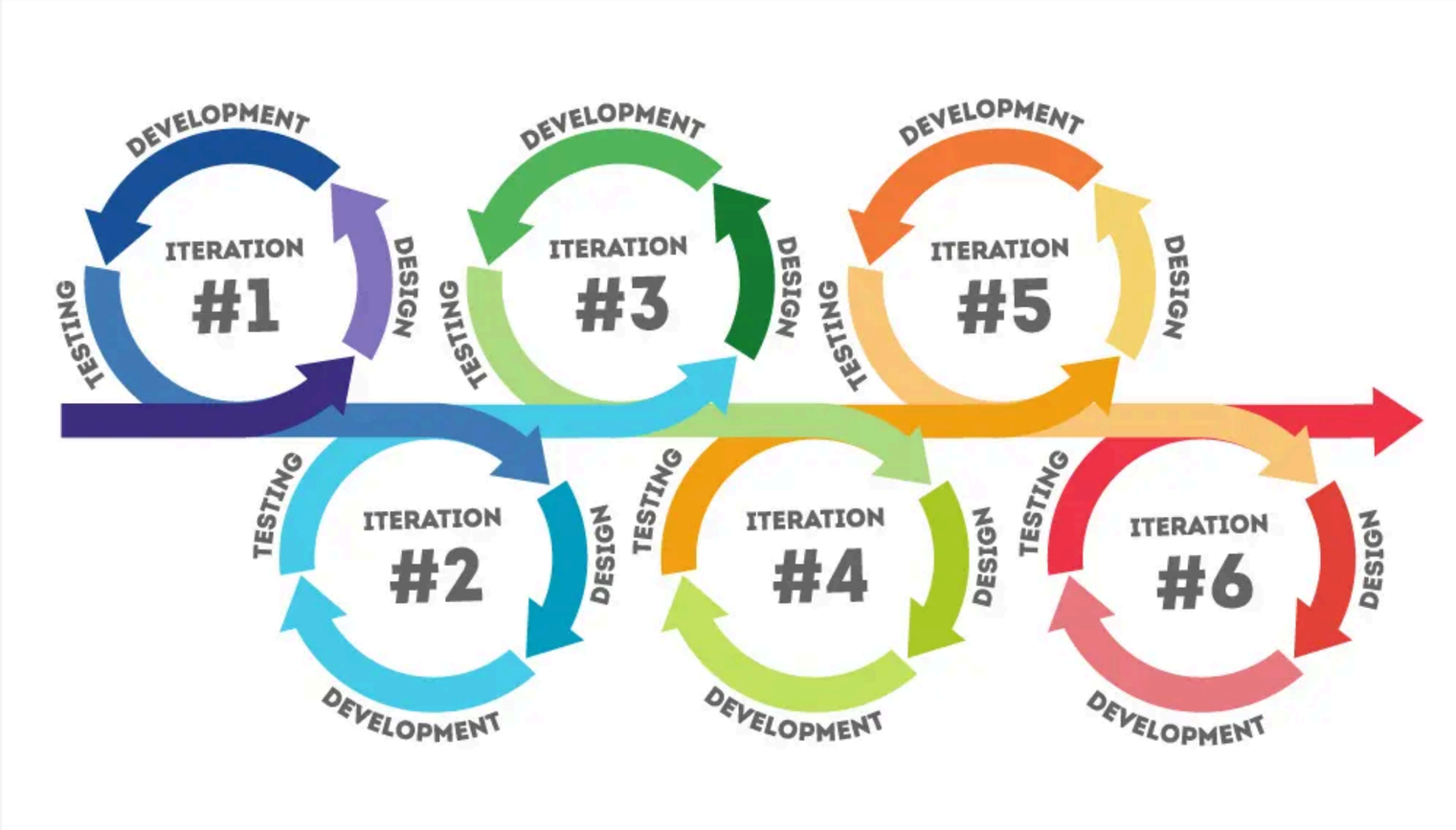
Car Rental example

Agile: The team agrees all car rental requirements, gathers feedback from branch staff, then adds reservations and payments in short iterations.

Waterfall: The team agrees all car rental requirements upfront, then completes design, implementation, testing and deployment in sequence.

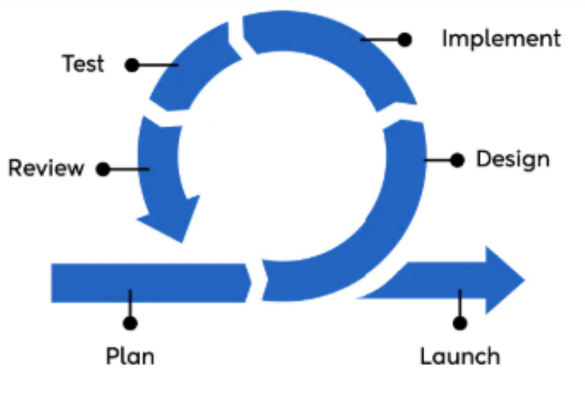
Iterative Development

The solution is refined through **repeated cycles.**



Iterative Development - Car Rental example

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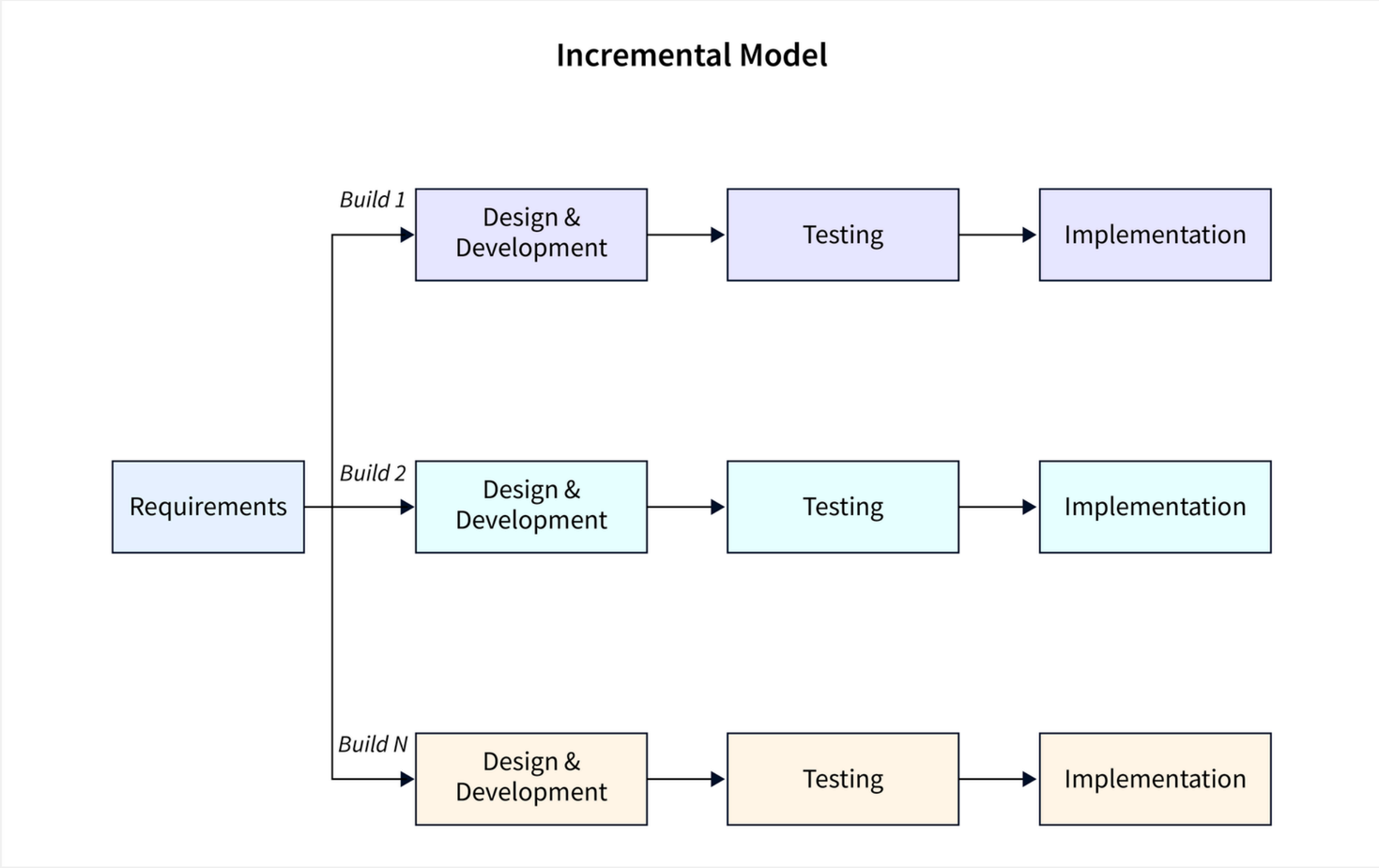


Car Rental example

First version of booking flow → branch staff review → improve cancellation handling.

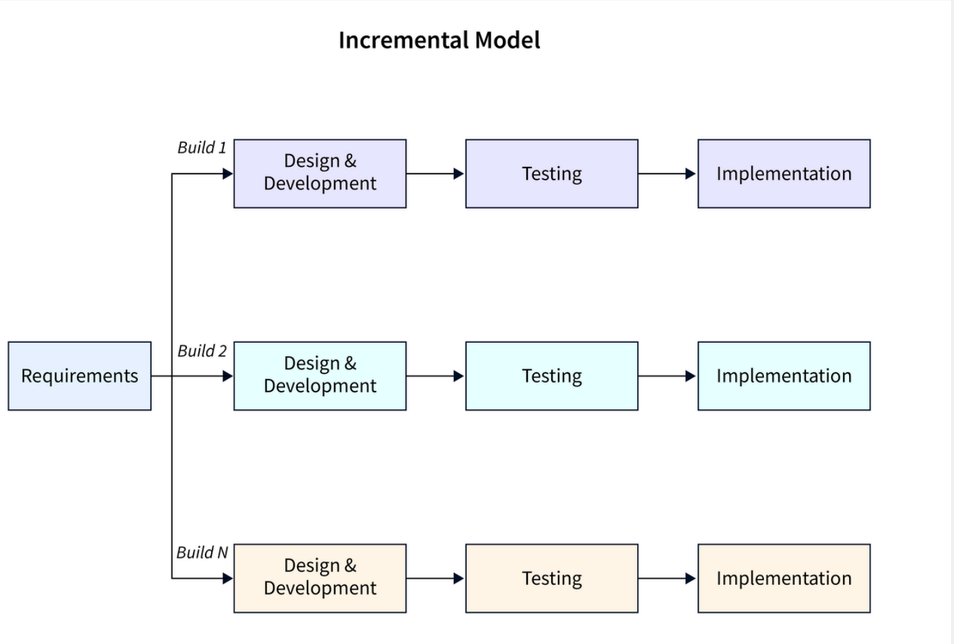
Incremental Development

The system is delivered in parts.



Incremental Development
Car Rental example

The system is delivered in parts.



Car Rental example:

1. Increment 1 → vehicle availability
2. Increment 2 → reservations
3. Increment 3 → payments
4. Increment 4 → staff functions

Iterative vs. Incremental Development

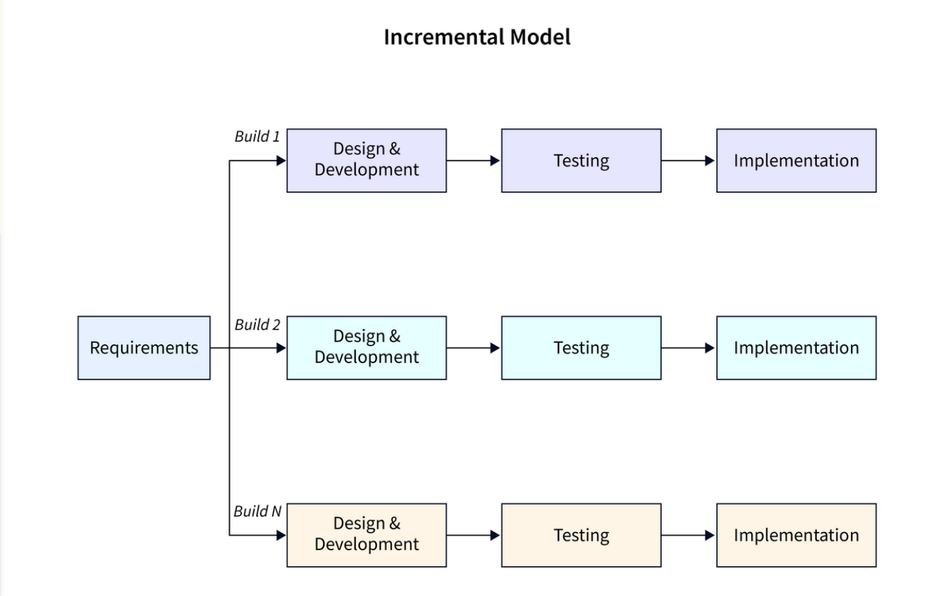
Iterative

Improve what already exists.

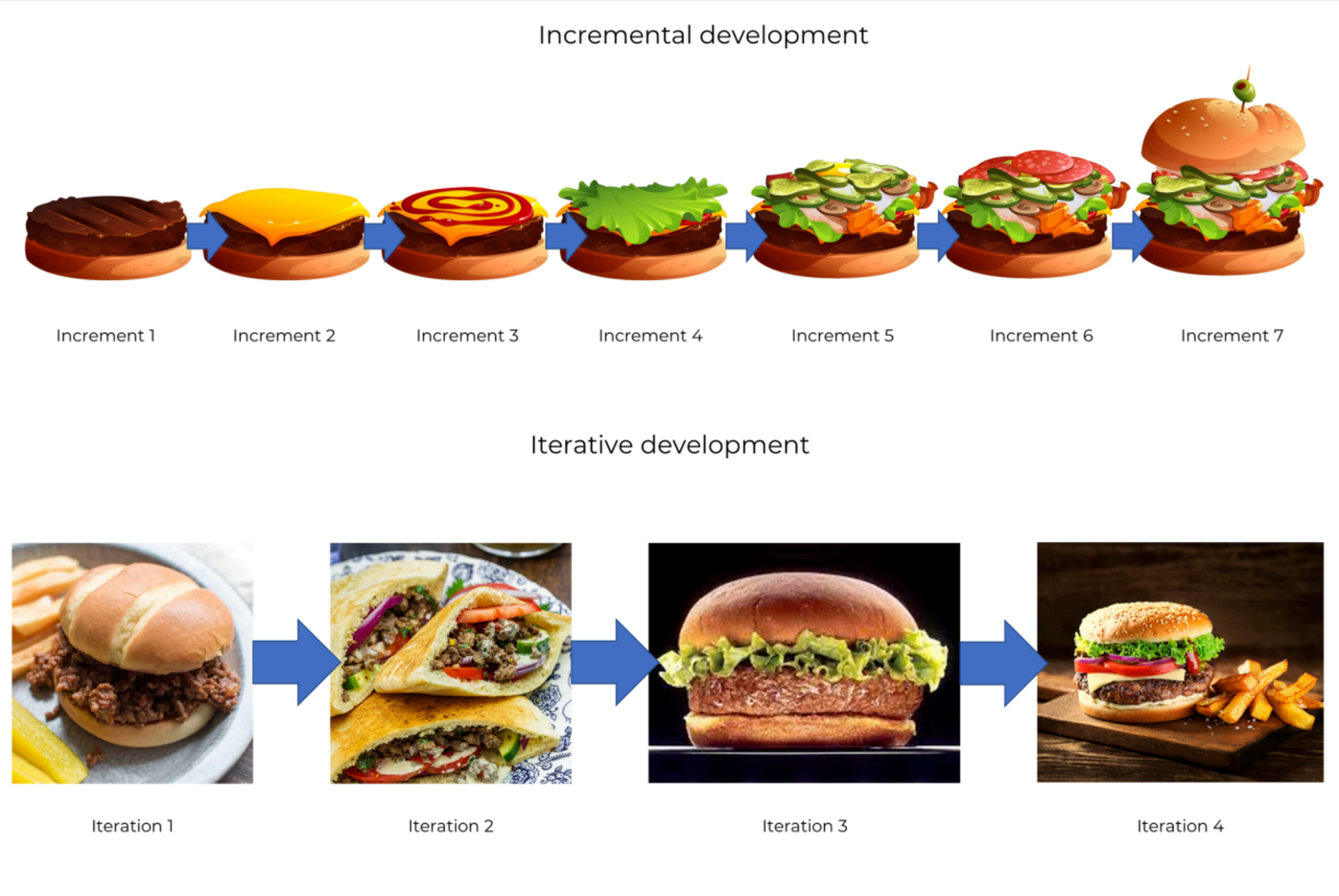


Incremental

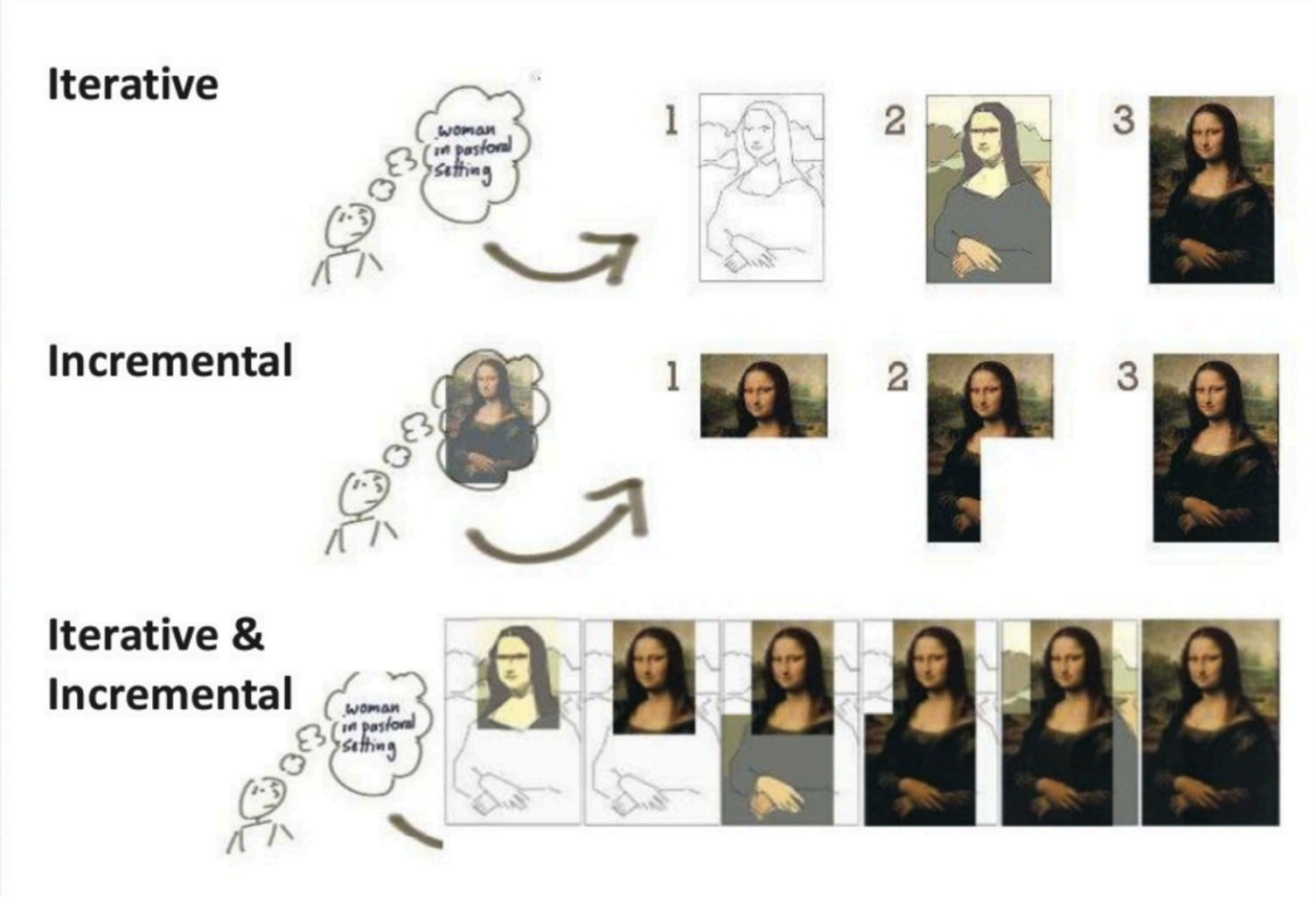
Add new functionality.



Iterative vs. Incremental Development



Iterative vs. Incremental Development



Iterative vs. Incremental Development

Car Rental example

Incremental

Add new functionality.

Iterative

Improve what already exists.

Car Rental example:

Improve reservation flow several times while gradually adding payment and staff tools.

Risk-driven development

High-risk uncertainties are investigated early.

YOU DO NOT SIMPLY BUILD THE EASIEST OR MOST VISIBLE FEATURES FIRST. YOU DELIBERATELY TACKLE THE PARTS THAT COULD SERIOUSLY THREATEN THE PROJECT.

High-risk assumptions should be tested early, before too much of the system depends on them.

Risk-driven development
Car Rental example

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YOU DO NOT SIMPLY BUILD THE EASIEST OR MOST VISIBLE FEATURES FIRST. YOU DELIBERATELY TACKLE THE PARTS THAT COULD SERIOUSLY THREATEN THE PROJECT.

Car Rental example:

Risk → The existing fleet system may not provide good availability data.

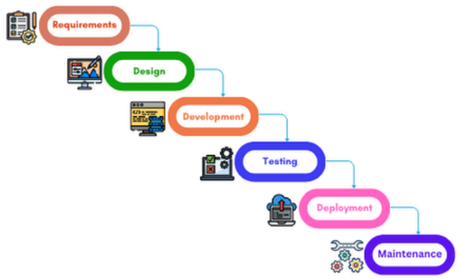
Early action → Prototype and test the integration first.

Why? → If availability cannot be trusted, the booking solution must be redesigned.

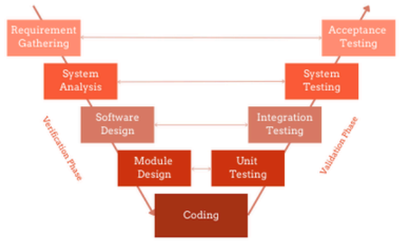
High-risk assumptions should be tested early, before too much of the system depends on them.

Choose What Fits Your Project's Needs

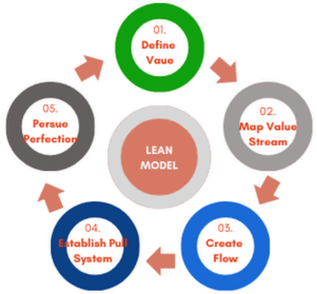
Not the One You Know Best



Waterfall Model



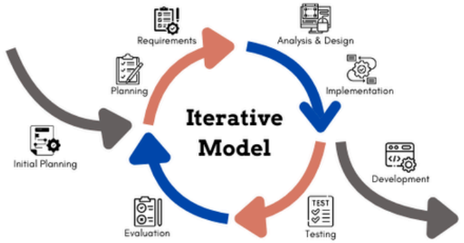
V Model



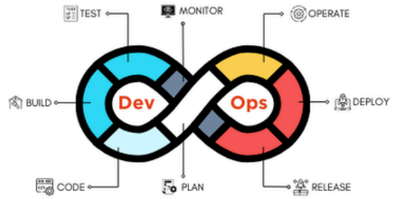
Lean Model



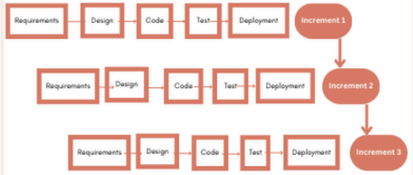
Spiral Model



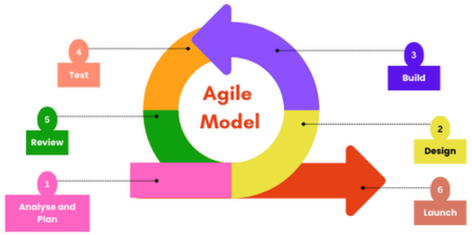
Iterative Model



DevOps Model



Incremental Model



Agile Model

Choose What Fits Your Project's Needs

Not the One You Know Best

Consider:

- **Requirements stability** — are requirements clear or likely to change?
- **Technical uncertainty** — are there unknown technologies or integrations?
- **Risk level** — which uncertainties could seriously affect the project?
- **Stakeholder availability** — can users give frequent feedback?
- **Delivery needs** — is early partial delivery useful?
- **Project size & complexity** — how difficult is coordination?
- **Documentation / regulation** — how much formal control is required?
- **Team experience** — which approach can the team realistically use?
- **Time constraints** — how fixed is the deadline?
- **Need for flexibility** — how easily must the project adapt to change?

Choose What Fits Your Project's Needs

Not the One You Know Best

Consider:

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Car Rental example:

If branch employees need to validate workflows early, a purely sequential approach may be risky.

Car Rental Lifecycle Decision

Possible reasoning:

Requirements are partly known, but staff workflow and system integration contain uncertainty.

Possible choice:

Iterative + incremental development.

Why?

- early feedback,
- gradual delivery,
- integration risks can be tested sooner.

Car Rental Lifecycle Decision

Possible reasoning:

Requirements are partly known, but staff workflow and system integration contain uncertainty.

Possible choice:

Iterative + incremental, with risk-driven priorities

Why?

- requirements are partly known,
- staff workflow needs feedback,
- functionality can be delivered gradually,
- risky integrations should be tested early.

What Is Feasibility?

Feasibility asks: Can this project realistically succeed?

Technical
Can we build the solution with available technology and skills?

Operational
Will the solution work in the organization?

Economic
Does expected value justify expected cost?

Schedule
Can it be delivered within the required time?

What Is Feasibility?

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Schedule
Can it be delivered within the required time?

Feasibility should consist of: assessment, assumption and uncertainty

Assessment — What is our current judgment? - A preliminary conclusion about whether the project is feasible in this dimension.

Assumption — What are we taking as true? - An unverified condition on which our assessment depends.

Uncertainty — What do we still need to check? - Missing information that could change our assessment.

What Is Feasibility?

Feasibility asks: Can this project realistically succeed?

Car Rental example: Can we build it, use it, afford it and deliver it in time?

Technical

Can we build the solution with available technology and skills?

Car Rental example

- **Assessment**: Payment integration appears technically feasible.
- **Assumption**: The provider offers an API compatible with our system.
- **Uncertainty**: Does the API support the required payment and refund operations?

Operational

Will the solution work in the organization?

Car Rental example

- **Assessment**: The new reservation workflow appears workable for branch staff.
- **Assumption**: Staff have time to update vehicle availability during daily work.
- **Uncertainty**: Can staff keep availability accurate during busy periods?

Economic

Does expected value justify expected cost?

Consider: development, infrastructure, training, maintenance, savings, business benefits.

Car Rental example

- **Assessment**: Savings from reduced manual work may justify the project's costs.
- **Assumption**: Online reservations will reduce staff time spent handling bookings.
- **Uncertainty**: Will those savings outweigh development, training and maintenance costs?

Schedule

Can it be delivered within the required time?

Car Rental example

- **Assessment:** A first version covering availability and reservations appears achievable within three months.
- **Assumption:** A suitably skilled team is available and the initial scope stays limited.
- **Uncertainty:** Can the necessary fleet integration be completed within that timeframe?

Feasibility Is a Decision

Possible outcomes:

GO

Proceed.

CONDITIONAL GO

Proceed if conditions are met.

NO-GO

Do not proceed in the current form.

Feasibility Is a Decision

Possible outcomes:

GO

Proceed.

Car Rental example: Fleet integration works, staff workflow is acceptable, budget and schedule are realistic.

CONDITIONAL GO

Proceed if conditions are met.

Car Rental example: Continue if the fleet integration prototype proves reliable.

NO-GO

Do not proceed in the current form.

Car Rental example: Existing systems cannot provide reliable availability data and replacement is outside budget.

HOW TO DECIDE?

Consider:

- technical feasibility,
- operational feasibility,
- economic feasibility,
- schedule feasibility,
- major unresolved risks.

GO = feasible overall
CONDITIONAL GO = feasible if key uncertainties are resolved
NO-GO = major blockers make the current proposal unrealistic

What Is a Risk?

A risk is an uncertain event or condition that may affect project objectives.

Common sources of risk:

- Unclear or changing requirements
- Unreliable external integrations
- Limited time, budget or skills
- Difficulties adopting a new workflow

What Is a Risk?

A risk is an uncertain event or condition that may affect project objectives.

Uncertainty → event → impact

Weak:

Integration problem.

Better:

The fleet system may not update vehicle status frequently enough, causing unavailable cars to appear bookable.

Risk Management Process

A risk is an uncertain event or condition that may affect project objectives.

Uncertainty → event → impact

Weak:

Integration problem.

Better:

The fleet system may not update vehicle status frequently enough, causing unavailable cars to appear bookable.

1. **Identify.**
2. **Analyze.**
3. **Plan response.**
4. **Monitor.**

Car Rental example

Risk → outdated fleet data

Response → prototype integration early

Risk Register

A risk register helps the team make risks explicit and decide **how they should be handled before they become real problems**. Probability describes how likely the risk is. Impact rates its seriousness. Consequence explains what happens if it occurs.

Car rental example

Risk: Fleet availability updates may be delayed.

Consequence: Customers may reserve cars that are unavailable.

Response: Test the integration early and agree an acceptable update delay.

A risk may happen. An issue has already happened.

Prototype vs. Pilot

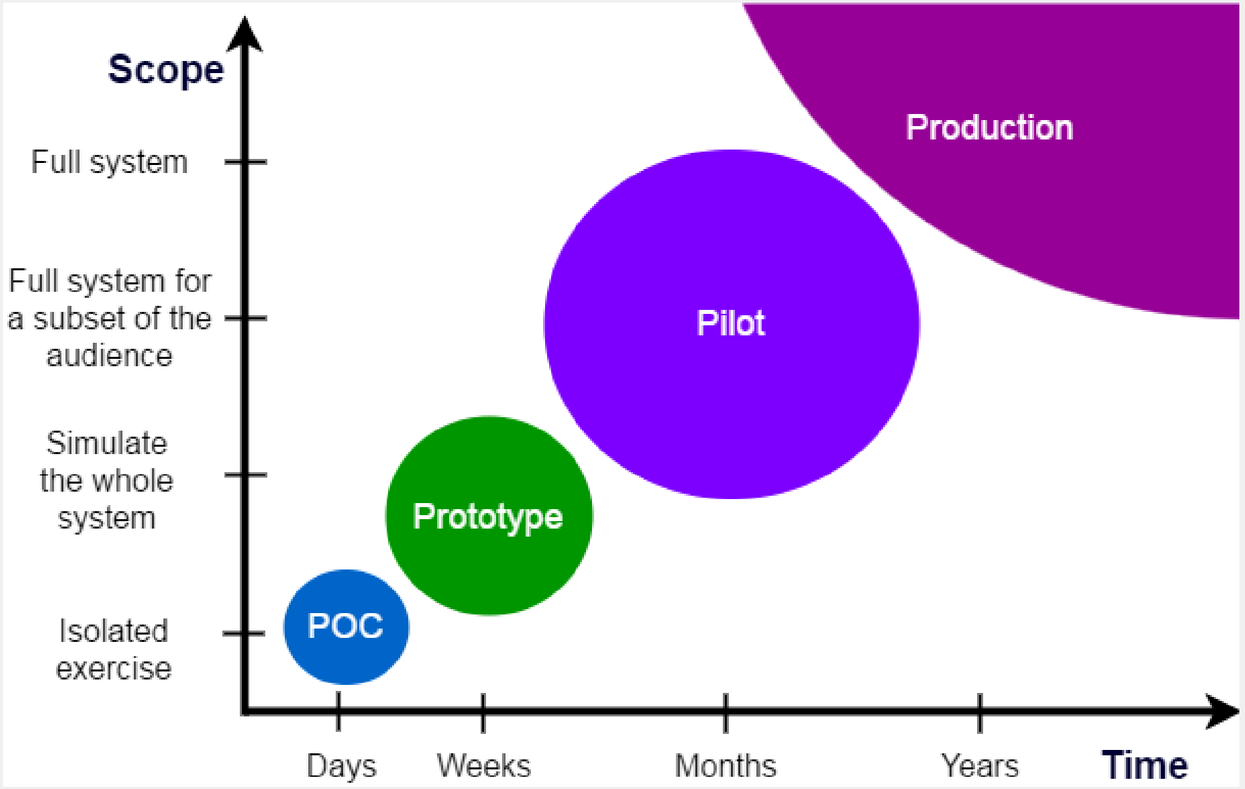
A prototype is an early, tangible model used to test an idea or design, while a pilot is a real-world, controlled trial of a nearly finished product or service

Prototype

Tests an idea or technical uncertainty.

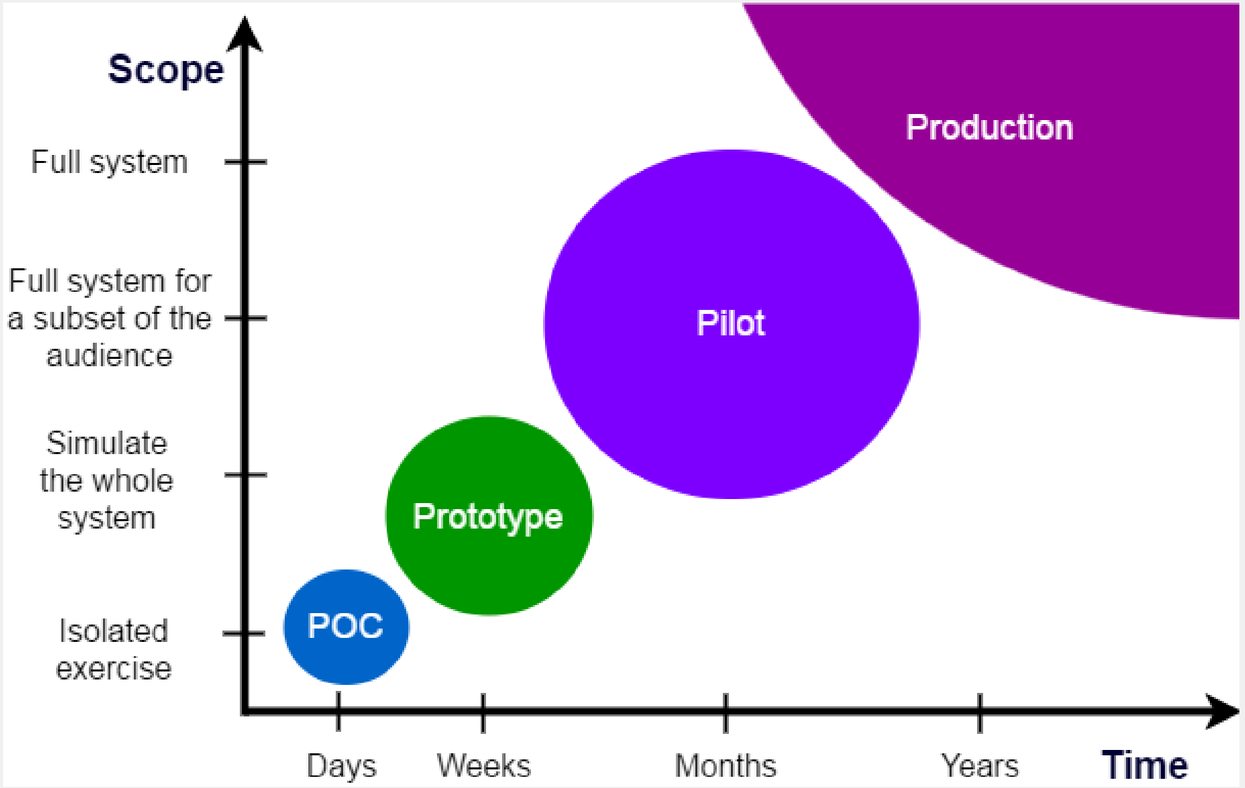
Pilot

Tests the solution in a limited real environment.



Prototype vs. Pilot

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Prototype

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Pilot

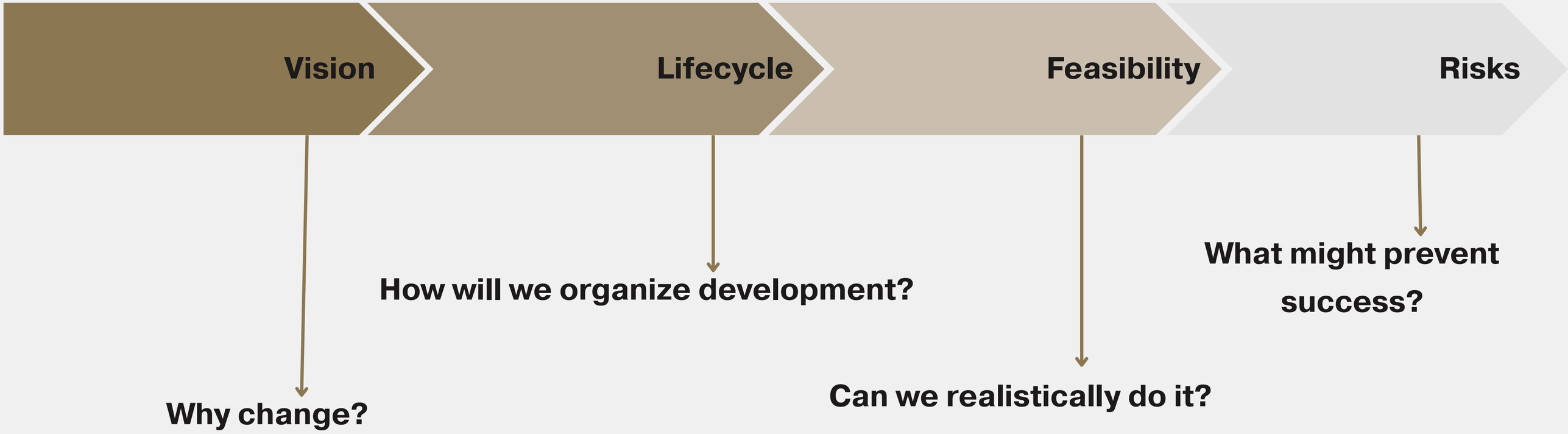
Tests the solution in a limited real environment.

Car Rental example

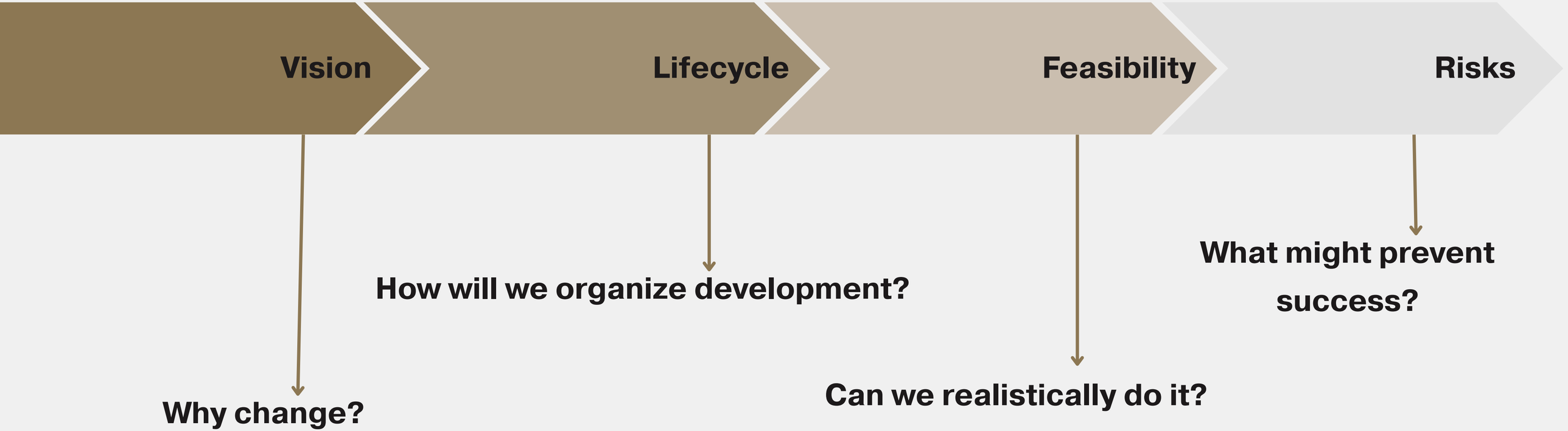
Prototype → test fleet integration

Pilot → launch in one branch first

So To Sum It Up...



So To Sum It Up…



Car Rental example

Reduce booking conflicts ➜ iterative approach ➜ conditional feasibility ➜ integration risk managed early.

Student Project: QuickBite

A fictional food-ordering and delivery platform connecting customers, restaurants and couriers

The website contains the full company brief, task instructions and required evidence.

M01 - Student Task - Teams of 3

Website module M01 - Tasks 6 & 8

Start / resume on the website

1. Open the same QuickBite website.
2. If needed, use Save & Export > Import project with your latest team ZIP.
3. Set Current session to 2; stay in website module M01.
4. Read M01 guidance and complete Tasks 6 & 8.

No starter ZIP, local server or Python installation is needed.

https://quickbite-workbench.vercel.app/

QuickBite: how to work, save and submit

***Work in a team of 3**. Use one agreed Team ID and produce one shared M01 project package. Discuss decisions together: each member should be able to explain the submitted work. The website contains the QuickBite company brief and task instructions; lecture slides explain theory using another company.*

1. Read the project information
Open Project Brief for the company background and project rules. Open M01 → Concepts / Student task for explanations, instructions and completion checklists.

2. Work directly on the website
Your workspace is in this browser. Write on task cards and click Save answer.
For Tasks 3–4, use Your project → Save stakeholders / Save scope.
Saved work is not shared automatically with teammates or the teacher.

3. Export before leaving
Open Save & Export → Save identity & download project ZIP.
Keep the ZIP outside the classroom computer.
Next class, open the same website and use
Save & Export → Import project with your latest ZIP.

M01:
4. Complete the required tasks
Session 1: Tasks 1–5. Session 2: Tasks 6 and 8.
Task 7 is optional: extra practice, no extra points and no penalty for skipping it.

5. M01 is worth 10 points in total
Five criteria, 0–2 points each.

M01 assessment 10 points total

Criterion	Points
Problem vision and explicit uncertainty	0–2
Stakeholders and conflict	0–2
Scope and two measurable objectives	0–2
Justified lifecycle choice	0–2
Feasibility three risks and decision	0–2

0 = absent or fundamentally incorrect;
1 = partly correct but incomplete or unclear;
2 = meets the checklist with clear, consistent reasoning. M01 is assessed as one module across both sessions. Optional exercises do not increase its maximum score. "Saved" confirms storage/structure, not correctness or a grade.

6. Submit one complete project ZIP on Moodle after each Checkpoint.
Do not edit the exported files manually or submit only M01.md. The ZIP contains your report (workspace/reports/M01.md), task answers and other data. Make corrections on the website, then export again.*
***If you want to update .md report file manually, copy it outside of the exported folder, then edit and upload additionally to Moddle's Module folder if needed.**

END OF CLASS

One website. One team project.
Keep your ZIP.

https://quickbite-workbench.vercel.app

Save your evidence

Reports/code save in the browser. Save models and simulator runs using their save buttons.

Download your project ZIP

Save & Export > Save identity & download project ZIP. Check Module deliverables.

Keep and share the agreed file

Store it outside the university computer. Submit on Moodle when requested; each backup is not automatically a submission.

Next class: return and import

Open the same website; import the latest ZIP if needed. Select the new Current session. Do not extract and start a server.

THANK YOU